

Experiment-4

To perform the function-oriented diagram: Data Flow Diagram (DFD) for Mail Defender.

1. Introduction to Function-Oriented Design and SA/SD Methodology

Function-Oriented Design is a traditional software engineering approach in which a complex system is decomposed into smaller functions. Each function receives input data, transforms it, and produces output data. This approach is useful when the main purpose of the system is to process information in a clear input-process-output manner.

The Structured Analysis and Structured Design methodology uses this idea to understand what the system does before deciding how it will be coded. During Structured Analysis, the problem description from the SRS is converted into logical diagrams such as Data Flow Diagrams. During Structured Design, these logical diagrams can later be converted into module-based structure charts.

For Mail Defender, this modeling method is highly suitable because the project receives suspicious email data and transforms it through parsing, IOC extraction, reputation enrichment, risk scoring, AI-assisted analysis, and report generation. The DFD helps the reader understand the data movement without needing to know the actual code or backend implementation details.

2. Comparing Flowcharts, DFDs, and Structure Charts

Feature	Flowchart	Data Flow Diagram	Structure Chart
Primary Focus	Step-by-step execution and decision sequences.	Logical movement and transformation of data.	Software hierarchy of software modules.
Control Logic	Shows loops, conditions, and procedural flow.	Does not focus on loops or control details.	Shows module calls and data/control couples.
Best Use	Algorithm explanation.	Requirement and analysis modeling.	Design and architecture modeling.

3. Data Flow Diagrams (DFD) in Detail

A Data Flow Diagram maps how information enters a system, how it is transformed by processes, where it is stored, and what outputs are sent to external entities. DFDs are logical diagrams; they describe what data moves and what transformations happen, not the exact programming sequence.

3.1 Standard DFD Symbols

- **Process:** Represents a function that transforms input data into output data. In Mail Defender, examples include Parse Email, Extract Evidence, Score Risk, and Build Report.
- **Data Flow:** Represents movement of data between external entities, processes, and data stores. Examples include Suspicious Email, Extracted IOCs, Reputation Data, Risk Verdict, AI Summary, and Final Report.
- **Data Store:** Represents a logical place where data is held for later use. In the current MVP this may be temporary state, while future versions may store analysis history and reports in a database.
- **External Entity:** Represents people or systems outside the Mail Defender boundary, such as User/Analyst, Threat Intelligence APIs, AI Analysis API, and Report Consumer.

3.2 Important DFD Rules Followed

- Every process is named using a verb-based function name, such as Parse Email or Build Report.
- Every data flow is labeled with the data being transferred, not with a control instruction.
- External entities do not directly access data stores; data passes through system processes.
- The Level 0 and Level 1 DFDs are balanced, meaning the major inputs and outputs shown at Level 0 are preserved after decomposition in Level 1.

3.3 Hierarchical DFD Levels

- **Level 0 Context DFD:** Shows the whole Mail Defender system as one process and focuses on the system boundary and external data exchange.
- **Level 1 DFD:** Breaks the central process into major sub-processes such as email intake, parsing, IOC extraction, enrichment, risk scoring, AI summary, and report generation.

4. Applied Case Study: Mail Defender

Mail Defender is a web-based suspicious email analysis platform. A user can upload an .eml file or paste raw email headers and body content. The system analyzes sender authentication, extracts suspicious indicators such as URLs, domains, IP addresses, and file hashes, enriches those indicators using threat intelligence APIs, calculates a phishing risk score, and generates a final report.

The system is function-oriented because its workflow can be divided into clear transformations: raw email input becomes parsed email data; parsed data becomes security evidence; evidence becomes enriched intelligence; enriched intelligence becomes risk score and verdict; and final verdict becomes a report.

4.1 Objective of the Mail Defender DFD

- To show how suspicious email information enters the system and moves through each processing stage.
- To identify the main external entities that exchange data with Mail Defender.
- To separate logical processes such as parsing, IOC extraction, enrichment, scoring, and reporting.
- To show temporary and future data stores such as raw email input, extracted evidence, and analysis result archive.

4.2 Main Inputs

- Raw email headers, email body, and uploaded .eml or .txt files.
- Attachment content and metadata, including filename, content type, size, and hash.
- External reputation responses from VirusTotal and AbuseIPDB.

4.3 Main Outputs

- Parsed header details such as From, Reply-To, Return-Path, Subject, Message-ID, and originating IP.
- Extracted indicators of compromise such as URLs, domains, IPs, email addresses, and hashes.
- Risk score, verdict, attack type, key findings, analyst summary, recommended actions, and PDF report.

5. Level 0 Context DFD for Mail Defender

The Level 0 Context DFD represents the complete Mail Defender system as a single process. It shows the main external entities and the primary data that crosses the system boundary. Internal processes and data stores are intentionally hidden at this level so that the reader can first understand the system at a high abstraction level.

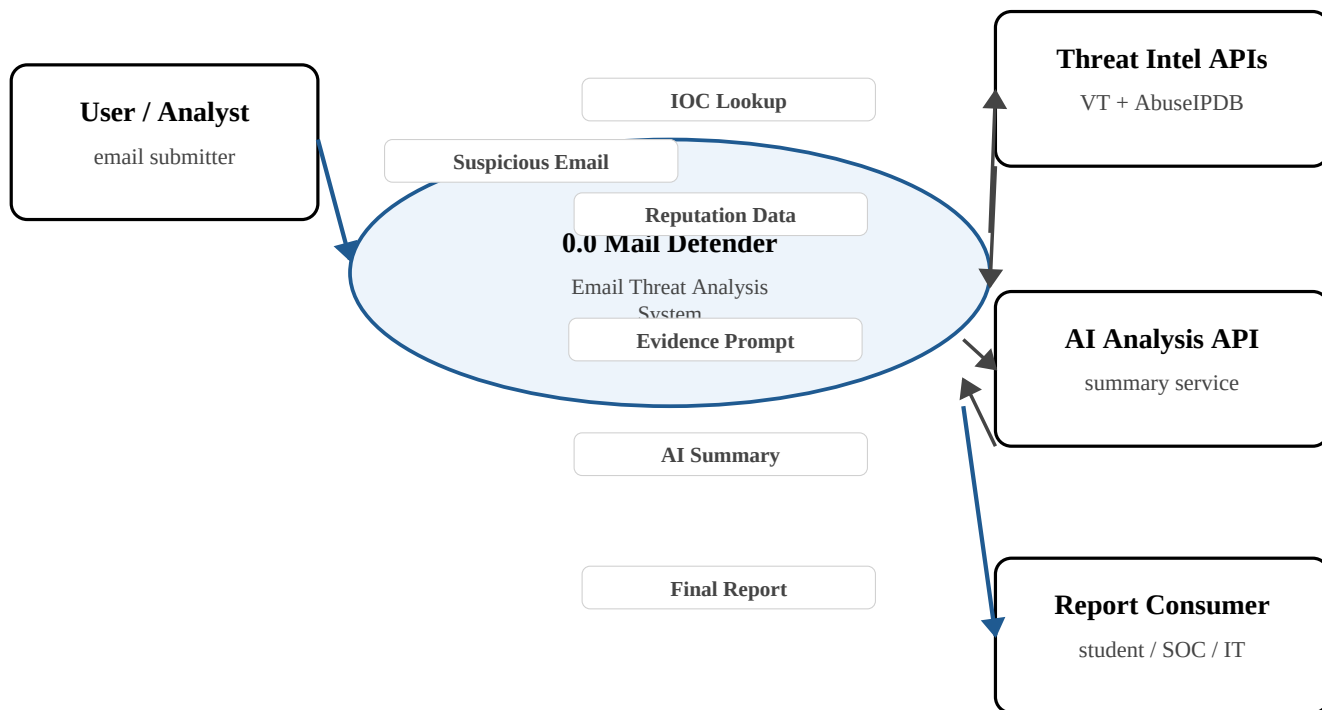


Figure 1: Level 0 Context DFD for Mail Defender

5.1 Explanation of Level 0 DFD

- The User/Analyst provides suspicious email data to Mail Defender. This can be an uploaded .eml file, a .txt file, or manually pasted headers and body.
- Mail Defender sends extracted indicators such as URLs, domains, IP addresses, and hashes to Threat Intelligence APIs for reputation checking.
- Threat Intelligence APIs return reputation data such as malicious, suspicious, clean, or unknown disposition.
- Mail Defender sends structured evidence to the AI Analysis API when AI support is configured.
- The AI Analysis API returns a human-readable analyst summary, key findings, and recommended actions.
- Finally, Mail Defender produces a security analysis report for the report consumer, which may be a student, IT administrator, or SOC analyst.

5.2 Level 0 Data Flow Summary

Data Flow	Source	Destination	Purpose
Suspicious Email	User / Analyst	Mail Defender	Provides raw email data for analysis.
IOC Lookup	Mail Defender	Threat Intel APIs	Checks URLs, domains, IPs, and hashes.
Reputation Data	Threat Intel APIs	Mail Defender	Returns malicious, suspicious, clean, or unknown evidence.
Evidence Prompt	Mail Defender	AI Analysis API	Sends structured evidence for summary generation.
AI Summary	AI Analysis API	Mail Defender	Returns analyst summary and recommendations.
Final Report	Mail Defender	Report Consumer	Provides final readable report.

6. Level 1 DFD for Mail Defender

The Level 1 DFD decomposes the central Mail Defender process into its major internal functions. This level shows how data flows between internal processes and data stores.

6.1 Level 1 DFD Diagram

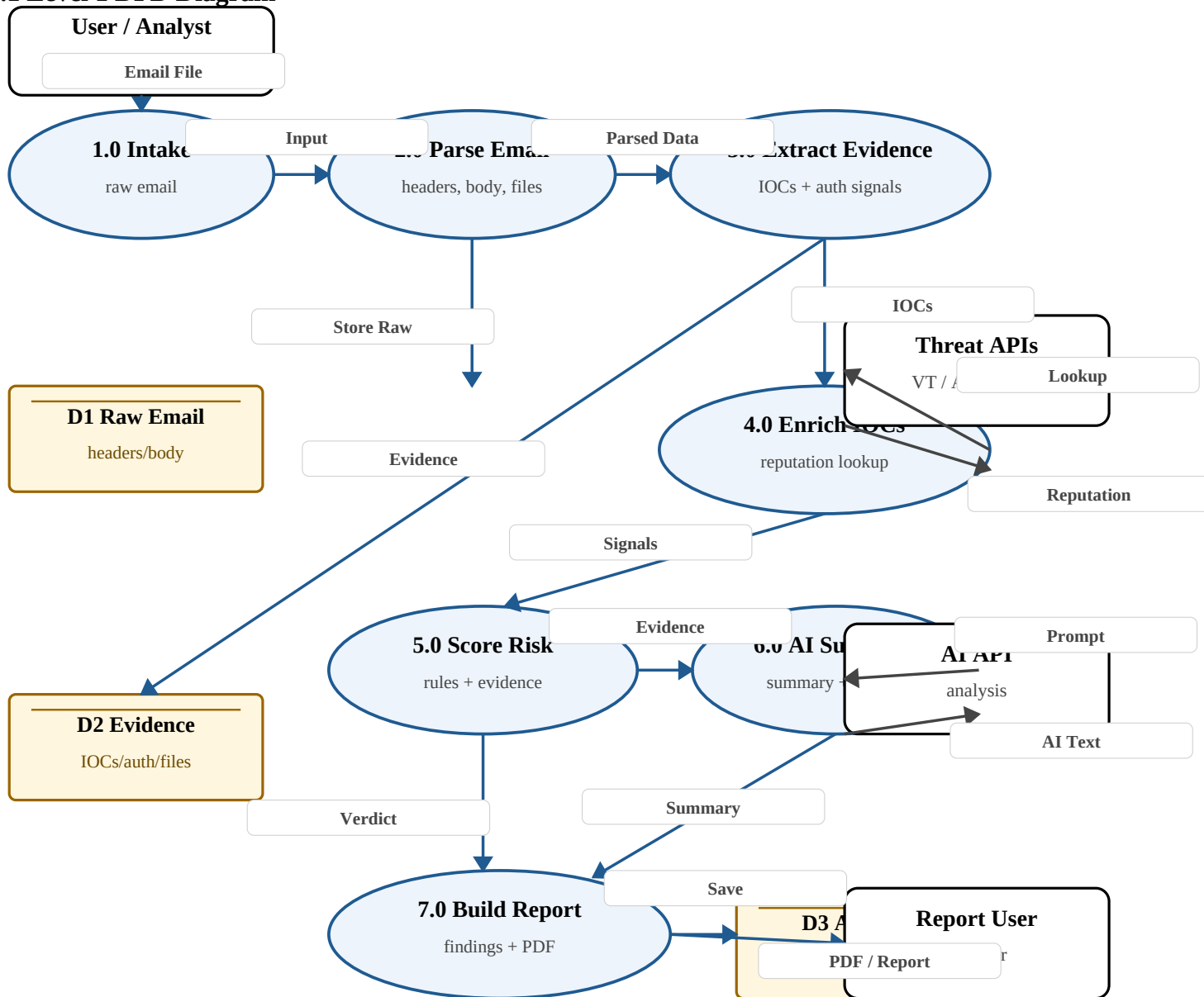


Figure 2: Level 1 DFD for Mail Defender

6.2 Explanation of Level 1 DFD

- **1.0 Email Intake:** Accepts the suspicious email from the user through upload or pasted input and performs basic validation.
- **2.0 Parse Email:** Splits the raw email into headers, body, HTML content, and attachments. Attachment hashes are calculated using SHA-256.
- **3.0 Extract Evidence:** Extracts IOCs such as URLs, domains, public IP addresses, emails, and hashes. It also reads SPF, DKIM, DMARC, reply-to mismatch, and spam status signals.
- **4.0 Enrich IOCs:** Sends indicators to external threat intelligence services such as VirusTotal and AbuseIPDB and receives reputation information.
- **5.0 Score Risk:** Applies local phishing rules and reputation evidence to generate a risk score, verdict, attack type, and risk factors.
- **6.0 AI Summary:** Uses structured evidence to generate a concise analyst summary and recommended actions.
- **7.0 Build Report:** Combines parsed details, evidence, enrichment, risk score, AI summary, and recommendations into a copyable or PDF security analysis report.

6.3 Process Specification Table

Process	Input Data	Output Data
1.0 Intake	Email file, pasted headers, pasted body	Validated input
2.0 Parse Email	Validated input	Parsed headers, body, attachments
3.0 Extract Evidence	Parsed email data	IOCs, auth signals, attachment hashes
4.0 Enrich IOCs	Extracted indicators	Reputation evidence
5.0 Score Risk	Signals and reputation evidence	Risk score and verdict
6.0 AI Summary	Structured evidence	Summary and recommended actions
7.0 Build Report	Verdict, evidence, AI text	Final PDF / report

7. Data Stores Used in the DFD

- **D1 Raw Email Input:** Temporarily stores raw headers, body content, and file data during analysis.
- **D2 Evidence:** Stores extracted IOCs, authentication results, attachment hashes, and local signals during the analysis workflow.
- **D3 Analysis Result:** Stores final verdict, score, attack type, findings, recommendations, and generated report information. In the current MVP this may be held in application state; in future company-level versions it can become a database-backed record.

7.1 Data Store Details

Data Store	Data Stored	Use in Mail Defender
D1 Raw Email Input	Headers, body, uploaded files	Used by Parse Email process to separate message parts.
D2 Evidence	IOCs, auth checks, attachment hashes	Used by enrichment and risk scoring processes.
D3 Analysis Result	Score, verdict, report text, recommendations	Used to create final report and future history.

8. DFD Balancing and Notes

The Level 1 DFD remains balanced with the Level 0 Context DFD. The suspicious email input shown in Level 0 is decomposed into email intake and parsing at Level 1. The IOC lookup and reputation response flows shown in Level 0 are represented through the Enrich IOCs process at Level 1. The AI summary flow is represented through the AI Summary process, and the final report output is represented through the Build Report process.

This DFD also separates current MVP behavior from future persistence. Current analysis can be performed in-memory during a user session, while future versions can permanently store raw input, extracted evidence, analysis results, report archive, and case management data.